

Database Design Assignment 1

Katie McGrath - R00261559

(a) What stage of development is the above table currently in within the design process: 1NF, 2NF, 3NF or BCNF? Explain your choice of answer. (5 Marks)

1NF → all attribute values are atomic, and there are no repeating groups. The data is flat, in 2D format with distinct columns and rows.

It is not in 2NF because it has partial dependencies.

(b) The data in the above table is susceptible to processing anomalies. Provide two examples for each of insertion, deletion, and update anomalies showing how they might occur in the above table. (15 Marks)

Insertion anomalies

(a) Adding a new car:

If a new car was to be assigned to an outlet, we cannot add it to the table unless the car has been hired at least once. The primary key would be incomplete without a *custNo* or *hireDate*, which would leave NULLs or fake data (invalid data).

(b) Adding a new customer:

If a new customer registers with the car company, we cannot add them to the table until they hire a car at least once. The primary key would be invalid without a *carReg* or *hireDate*.

Deletion anomalies

(a) Deleting the only hire for a car:

If a customer returns a car, and its hire record/data was deleted, we would lose all information about the car. This is due to the data being stored *only* with the hire record. We would lose its make, model and home outlet. Its whole existence would be deleted from the table.

(b) Deleting the last hire from an outlet:

If the last hire record for the last car based at one of the outlets is deleted, we lose all information about that outlet.

Update anomalies

(a) Changing a car's model:

If a car's model was inputted into the table incorrectly, we would have to update every row where the car's corresponding *carReg* appears, to correct it. This is to prevent data from becoming inconsistent - if a row was missed, the same car would have two different models in the database.

(b) Changing a customer's name:

If a customer's name changed, we would need to update every row where the customer's corresponding *custNo* appears. If a row was missed, the same customer would have two different names in the system. This could lead to problems with billing, reporting, and correspondence.

(c) Using the process of Normalization, carry out the transformation of the above table to BCNF. All the steps should be clearly shown and labelled; all keys and functional dependencies should be identified at each stage. Identify the primary and foreign keys in your final BCNF relations. (60 Marks)

Initial Functional Dependencies

The full functional dependency from the original table is:

- $(carReg, hireDate) \rightarrow make, model, outletNo, outletLoc, custNo, custName$

To uniquely identify any single row, you need both the car registration and the hire date.

This identifies smaller functional dependencies in the table:

- $carReg \rightarrow make, model, outletNo, outletLoc$

A car's unique registration number determines its make, model and the outlet it belongs to.

- $outletNo \rightarrow outletLoc$

An outlet's number determines its location.

- $custNo \rightarrow custName$

Converting the Table to 2NF (Removing the Partial Dependencies)

By removing the partial dependencies from the key of $(carReg, hireDate)$:

1. **Cars** → Contains attributes that depend solely on $carReg$.
 - FD: $carReg \rightarrow make, model, outletNo, outletLoc$
 - Candidate Key: $carReg$
2. **Customers** → Contains attributes that depend solely on $custNo$.
 - FD: $custNo \rightarrow custName$
 - Candidate Key: $custNo$
3. **Hires** → Contains the attributes of the hire itself.
 - FD: $carReg, hireDate \rightarrow custNo$
 - Candidate Key: $(carReg, hireDate)$

This results in 3 2NF Tables:

Table: Cars

carReg	make	model	outletNo	outletLoc
M565 0GD	Ford	Escort	01	Bandon
N743 TPR	Nissan	Sunny	01	Bandon
M134 BRP	Ford	Escort	02	Mallow
M611 0PQ	Nissan	Sunny	02	Mallow

→ **Primary Key (PK):** *carReg*

Table: Customers

custNo	custName
C100	Smith, J
C201	Hen, P
C313	Blatt, O
C295	Pen, T

→ **Primary Key (PK):** *custNo*

Table: Hires

carReg	hireDate	custNo
M565 0GD	14/5/98	C100
M565 0GD	15/5/98	C201
N743 TPR	16/5/98	C100
M134 BRP	14/5/98	C313
M134 BRP	20/5/98	C100
M611 0PQ	20/5/98	C295

→ **Primary Key (PK):** (*carReg*, *hireDate*)

→ **Foreign Key (FK):** *carReg* references *Cars(carReg)*

→ **Foreign Key (FK):** *custNo* references *Customers(custNo)*

Converting to 3NF (Removing Transitive Dependencies)

Both the *Customers* and *Hires* tables have no transitive dependencies. However, in *Cars*, we have the FD *carReg* → *outletNo*, *outletLoc*, but we also have *outletNo* → *outletLoc*. This means that *outletLoc* is transitively dependent on *carReg* through *outletNo*.

We need to make a new table for *Outlets* to get rid of this dependency.

1. **Outlets** → Contains attributes that depend solely on *outletNo*.
 - FD: *outletNo* → *outletLoc*
 - Candidate Key: *outletNo*
2. **Cars (Newly Modified)** → Now only contains *carReg*, *make*, *model* and *outletNo*.
 - FD: *carReg* → *make*, *model*, *outletNo*
 - Candidate Key: *carReg*.

This results in 2 3NF Tables:

Table: Outlets

outletNo	outletLoc
01	Bandon
02	Mallow

→ **Primary Key (PK):** *outletNo*

Table: Cars (Newly Modified)

carReg	Make	Model	outletNo
M565 0GD	Ford	Escort	01
N743 TPR	Nissan	Sunny	01
M134 BRP	Ford	Escort	02
M611 0PQ	Nissan	Sunny	02

→ **Primary Key (PK):** *carReg*

→ **Foreign Key (FK):** *outletNo* references *Outlets(outletNo)*

Convert to BCNF

All the tables are already in BCNF form:

- **Outlets:** *outletNo* is the primary key and superkey.
- **Customers:** *custNo* is the primary key and superkey.
- **Cars:** *carReg* is the primary key and superkey.
- **Hires:** (*carReg*, *hireDate*) is the primary key and superkey.

This is due to fixing the transitive dependency that invalidated 3NF.

Final BCNF Schema (With PKs and FKs)

Table: Outlets

outletNo	outletLoc
01	Bandon
02	Mallow

→ **Primary Key (PK):** *outletNo*

→ **Attributes:** *outletLoc*

Table: Cars (Newly Modified)

carReg	Make	Model	outletNo
M565 0GD	Ford	Escort	01
N743 TPR	Nissan	Sunny	01
M134 BRP	Ford	Escort	02
M611 0PQ	Nissan	Sunny	02

→ **Primary Key (PK):** *carReg*

→ **Foreign Key (FK):** *outletNo* references *Outlets(outletNo)*

→ **Attributes:** *make, model, outletNo*

Table: Customers

custNo	custName
C100	Smith, J
C201	Hen, P
C313	Blatt, O
C295	Pen, T

→ **Primary Key (PK):** *custNo*

→ **Attributes:** *custName*

Table: Hires

carReg	hireDate	custNo
M565 0GD	14/5/98	C100
M565 0GD	15/5/98	C201
N743 TPR	16/5/98	C100
M134 BRP	14/5/98	C313
M134 BRP	20/5/98	C100
M611 0PQ	20/5/98	C295

→ **Primary Key (PK):** (*carReg, hireDate*)

→ **Foreign Key (FK):** *carReg* references *Cars(carReg)*

→ **Foreign Key (FK):** *custNo* references *Customers(custNo)*

→ **Attributes:** *custNo*

(d) Draw an Entity-Relationship model for the normalization process completed in step (c) above. Show all the entities, relationships, and attributes. (20 Marks)

